

## FINTRUST DIGITAL BANK

### Final Business Intelligence & Dashboard Solution

#### Week 4 Final Project Report – Data Analytics Track



|                      |                                                         |
|----------------------|---------------------------------------------------------|
| <b>Prepared by</b>   | Joyce Wambui Kibandi                                    |
| <b>Programme</b>     | AnalystLab Africa – Experience Lab Internship Programme |
| <b>Project stage</b> | Week 4 – Test, Refine & Present                         |
| <b>Date</b>          | 10 <sup>th</sup> October ,2026                          |

*Final-stage focus: Test → Validate → Refine → Finalize → Present*

## Report Structure

1. Executive Summary
2. Project Background and Business Objective
3. Data Sources and Final Data Quality Status
4. Analytical Approach and Tools
5. Weeks 1–3 Foundation and Week 4 Refinement
6. Final SQL Analysis
7. Final Python Analysis and Reproducibility Review
8. Final Power BI Dashboard
9. Final KPI Analysis
10. Final Business Insights and Recommendations
11. Testing, Validation and Refinement Evidence
12. Integration with the Wider FinTrust Solution
13. Assumptions, Limitations and Responsible Use
14. Lessons Learned and Final Reflection
15. Final Deliverables and Conclusion

### 1. Executive Summary

**Purpose.** This report presents the final Data Analytics solution developed for the fictional FinTrust Digital Bank project. It consolidates the work completed across Weeks 1–4 and demonstrates how customer and transaction data were prepared, analysed, validated, visualised and translated into decision-oriented business intelligence.

**Final outcome.** The final solution combines SQL, Python and Power BI. SQL was used for business-oriented querying and comparative analysis; Python was used for exploratory analysis, cross-validation and reproducibility checks; and Power BI was used to communicate KPIs, transaction patterns, customer behaviour, risk-review patterns and drill-through detail in an interactive dashboard.

- 1,500 customer records and 12,000 transaction records were retained in the final cleaned analytical datasets.
- Core KPIs were independently recalculated and matched the values used in the final dashboard.
- The strongest analytical findings were cross-validated across SQL, Python and Power BI.
- Week 4 testing found no material inconsistencies requiring changes to the underlying cleaned datasets or major analytical calculations.
- Final refinements focused on validation evidence, dashboard clarity, drill-through functionality, KPI consistency and final documentation.

### 2. Project Background and Business Objective

FinTrust is a fictional digital banking organisation used for the AnalystLab Africa Experience Lab project. The Data Analytics track focused on transforming the supplied customer and transaction data into a coherent business intelligence solution that can support management understanding and decision-making.

**Primary objective:** Provide reliable, interpretable and decision-oriented analysis of customer behaviour, transaction performance, channel activity, operational failures, digital engagement and risk-review patterns.

**Final Week 4 objective:** Move from development to final testing, validation, refinement, documentation and presentation of the analytical solution.

### 3. Data Sources and Final Data Quality Status

| Dataset          | Final size              | Purpose                                                                                                            |
|------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------|
| Customer data    | 1,500 rows / 12 fields  | Customer profile, segment, account, tenure, digital engagement, income band, preferred channel and account status. |
| Transaction data | 12,000 rows / 12 fields | Transaction date/time, type, value, channel, device, location, international flag, status and risk-review flag.    |

- Customer\_ID is the analytical key used to connect the customer and transaction datasets.
- Customer\_Name is not treated as a unique identifier because names can repeat across customers.
- The final transaction period covers 1 January 2026 to 31 March 2026.
- Transaction\_ID and Customer\_ID uniqueness checks passed for their intended roles.
- All transaction Customer\_ID values matched valid customers in the customer dataset.
- No zero or negative transaction amounts were identified in the final dataset.
- Previously identified missing Device\_Type and Location values were handled during data preparation by assigning an explicit N/A category rather than silently dropping records.

**Final data-quality conclusion:** The cleaned datasets remained structurally consistent and suitable for final analysis. Week 4 checks did not identify a new material data-quality issue that required reopening the cleaning stage.

### 4. Analytical Approach and Tools

| Tool / method           | Role in the final solution                                                                                                           |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| MySQL / SQL             | Business querying, aggregation, segmentation, rankings, month-over-month comparison, failure-rate analysis and risk-review analysis. |
| Python / Pandas / NumPy | Data-quality checks, reproducibility testing, cross-validation, exploratory analysis and analytical summaries.                       |
| Matplotlib / Seaborn    | Supporting visual analysis, including trend charts, distributions and failure-rate heatmaps.                                         |
| Power BI                | Final KPI reporting, transaction/customer/risk analysis, slicers, cross-filtering and customer-segment drill-through.                |
| Git / GitHub            | Portfolio organisation, weekly documentation, code/notebook storage and project progression tracking.                                |

## 5. Weeks 1–3 Foundation and Week 4 Refinement

| Stage                           | Main contribution                                                                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1 – Understand & Plan      | Defined the business problem, analytical questions, KPIs, dashboard direction, risks, success criteria and project plan.                                                                  |
| Week 2 – Analyse & prepare      | Completed data-quality assessment, cleaning, descriptive SQL, Python EDA, KPI development and the initial Power BI dashboard.                                                             |
| Week 3 – Develop & Integrate    | Added advanced SQL, deeper Python analysis, normalized customer metrics, risk/failure analysis, cross-tool validation, management recommendations and interactive Power BI drill-through. |
| Week 4 – Test, Refine & Present | Rechecked data quality, recalculated KPIs, verified key SQL results in Python, tested notebook reproducibility, audited Power BI functionality and consolidated the final documentation.  |

## 6. Final SQL Analysis

The final SQL deliverable retains the most decision-relevant queries rather than adding unnecessary new analyses. The final set demonstrates JOINS, GROUP BY, CASE expressions, CTEs, aggregations and window functions where appropriate.

| Final business query             | Technique emphasis           | What it reveals                                                                                          |
|----------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------|
| Monthly transaction performance  | CTE + LAG + aggregation      | February declined in volume and value; March recovered strongly and generated the highest monthly value. |
| Customer segment performance     | JOIN + CTE + normalized KPIs | SME leads transaction value per customer; Student leads transaction frequency per customer.              |
| Transaction failure hotspots     | GROUP BY + CASE              | POS Bill Payment and USSD Transfer show the highest observed failure rates.                              |
| Risk review by transaction value | CTE + CASE + aggregation     | Risk-review rate rises with transaction value and reaches 40.54% for NGN 200K+.                          |

## 7. Final Python Analysis and Reproducibility Review

Python was used both as an analytical tool and as an independent validation layer. Week 3 extended the exploratory analysis, while Week 4 focused on reproducibility and verification of the final results.

- Monthly transaction trend analysis validated the January–March totals used in SQL and Power BI.
- A channel × transaction-type failure-rate heatmap reproduced the SQL failure-rate pattern.
- Customer-segment transaction-value distributions highlighted substantial within-segment variation and high-value outliers.
- Risk-review analysis by amount band reproduced the same percentages observed in SQL and Power BI.

- The Week 4 notebook was restarted and run from beginning to end using the final cleaned datasets to confirm reproducibility.

**Reproducibility result:** Passed. The final Python checks executed successfully and reproduced the expected analytical outputs.

## 8. Final Power BI Dashboard

The final Power BI dashboard develops the Week 2 version into a more decision-oriented and interactive solution. Week 4 review focused on KPI accuracy, filters, titles, labels, visual consistency, drill-through and overall user experience.

| Dashboard page                   | Purpose / final content                                                                                |
|----------------------------------|--------------------------------------------------------------------------------------------------------|
| Executive Overview / KPI Section | Core customer, transaction, value, success and risk-review KPIs with slicers.                          |
| Transaction Analysis             | Transaction value by channel, transaction status, normalized segment value and monthly trend analysis. |
| Risk & Operations                | Risk-review rate by channel, transaction type and transaction amount band.                             |
| Customer & Segment Analysis      | Transactions per customer by segment.                                                                  |

**Final dashboard review:** KPI accuracy, slicers, cross-filtering, titles, labels, formatting and user experience were tested and passed after minor presentation refinements.

## 9. Final KPI Analysis

| KPI                       | Final value  | Interpretation                                                          |
|---------------------------|--------------|-------------------------------------------------------------------------|
| Total Customers           | 1,500        | Final active customer population represented in the customer dataset.   |
| Total Transactions        | 12,000       | Total transactions analysed for January–March 2026.                     |
| Total Transaction Value   | NGN 560.48M  | Aggregate transaction value across the final transaction dataset.       |
| Average Transaction Value | ≈ NGN 46,706 | Average transaction amount across all analysed transactions.            |
| Transaction Success Rate  | 90.47%       | Share of transactions with Successful status.                           |
| Risk Review Rate          | 19.60%       | Share of transactions selected for the synthetic risk-review indicator. |

Week 3 and Week 4 analysis also introduced normalized and diagnostic KPIs such as transaction value per customer, transactions per customer, failure rates by channel/transaction type, month-over-month changes and risk-review rates by amount band. These provide more decision value than relying only on overall totals.

## 10. Final Business Insights and Recommendations

| Finding                                 | Evidence                                                                                                                                                   | Recommended action                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Monthly performance                     | January recorded 4,133 transactions worth NGN 188.49M; February fell to 3,734 and NGN 175.79M; March recovered to 4,133 and NGN 196.20M.                   | Monitor month-over-month changes and investigate material movements by channel, transaction type and segment.   |
| High-value risk-review concentration    | Risk-review rate increases from about 17–22% in lower amount bands to 40.54% for transactions of NGN 200K and above.                                       | Maintain enhanced review attention for high-value transactions while periodically validating review thresholds. |
| Operational failure hotspots            | POS Bill Payment recorded a 7.22% failure rate and USSD Transfer 7.20%; Mobile App Transfer generated 93 failures because of high volume.                  | Monitor both failure percentage and absolute failure count when prioritizing operational investigation.         |
| Segment performance after normalization | SME generated the highest transaction value per customer (NGN 387,923.02), while Student customers had the highest frequency (8.23 transactions/customer). | Use normalized segment KPIs alongside totals when evaluating segment performance.                               |

## 11. Testing, Validation and Refinement Evidence

Week 4 followed the required final-stage logic: Final Output → Test → Finding → Refinement → Re-test → Final Result. The strongest checks are summarised below.

| Test area                     | Finding                                                                                                 | Refinement / action                                              | Final result |
|-------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------|
| Dataset structure and quality | Expected row counts, unique IDs, no unresolved nulls, valid referential integrity and positive amounts. | No structural correction required; final checks were documented. | Passed       |
| Core KPIs                     | Python recalculation matched the dashboard values.                                                      | Formatting reviewed for consistency.                             | Passed       |
| Monthly SQL result            | Python reproduced the same January–March totals.                                                        | No analytical correction required.                               | Passed       |
| Risk-review amount bands      | SQL, Python and Power BI produced matching rates, including 40.54% for NGN 200K+.                       | Amount-band ordering and presentation were reviewed.             | Passed       |
| Failure-rate analysis         | SQL, Python and Power BI showed the same main channel × transaction-type hotspots.                      | Matrix presentation improved for comparison.                     | Passed       |

| Test area                         | Finding                                                                                  | Refinement / action             | Final result |
|-----------------------------------|------------------------------------------------------------------------------------------|---------------------------------|--------------|
| Python reproducibility            | Notebook reran from cleaned datasets without analytical errors.                          | Execution order reviewed.       | Passed       |
| Power BI slicers and interactions | Relevant visuals responded correctly and overall values restored after clearing filters. | Interaction behaviour reviewed. | Passed       |

## 12. Integration with the Wider FinTrust Solution

The Data Analytics output contributes to the wider FinTrust solution by converting customer and transaction data into validated KPIs, business findings and operational patterns. These outputs are positioned upstream of predictive risk intelligence and other technical components in the overall solution flow.

| Analytics contribution                           | Potential wider use                                                             |
|--------------------------------------------------|---------------------------------------------------------------------------------|
| Cleaned customer and transaction data            | Reliable foundation for downstream analysis or modelling.                       |
| Validated KPIs and customer/transaction patterns | Shared business context for predictive or service-oriented components.          |
| Risk-review and high-value transaction patterns  | Context for risk-oriented analysis, subject to the synthetic-target limitation. |
| Operational failure hotspots                     | Supports discussion of service reliability and potential operational follow-up. |
| Interactive Power BI dashboard                   | Management-facing presentation layer for the Data Analytics contribution.       |

## 13. Assumptions, Limitations and Responsible Use

- FinTrust is a fictional organisation and the supplied datasets are synthetic.
- The analysis period is limited to January–March 2026; the observed trends should not be treated as long-term seasonality or forecasting evidence.
- The analytical solution is descriptive and diagnostic; it does not establish causality.
- Risk\_Review\_Flag is a synthetic educational indicator and is not evidence of actual fraud, financial crime or customer misconduct.
- The dashboard and analysis are educational prototypes and are not a production banking system.
- Recommendations are based on the available synthetic data and should be validated against real operational processes, policies and risk thresholds before any real-world use.

## 14. Lessons Learned and Final Reflection

### 14.1 What worked well

- Using SQL, Python and Power BI together created a strong cross-validation workflow rather than relying on a single tool.
- Normalized KPIs such as value per customer and transactions per customer revealed insights that overall totals alone could hide.
- Building Week 3 and Week 4 on top of earlier work preserved continuity and demonstrated progressive improvement rather than restarting the project.

### 14.2 Challenges encountered

- Maintaining consistent field names, file locations and notebook environments across multiple project weeks required careful organisation.
- Balancing analytical depth with dashboard readability required removing or replacing less informative visuals instead of continually adding more charts.
- Testing filters and formatting in Power BI required iterative review before the final dashboard behaved as intended.

### 14.3 Key lessons learned

- Validation is stronger when the same finding is reproduced independently across different tools.
- Business intelligence should distinguish overall totals from normalized performance and operational impact.
- A final analytical deliverable is not complete when the calculations work; it also needs reproducibility, clear documentation, limitations and a usable presentation layer.

## 15. Final Deliverables and Conclusion

| Required final Data Analytics deliverable | Final project output                                                                               |
|-------------------------------------------|----------------------------------------------------------------------------------------------------|
| Final Power BI dashboard                  | Final refined Power BI dashboard with KPI, transaction, customer, risk and drill-through analysis. |
| Final SQL scripts                         | Final SQL file containing the strongest business queries and advanced SQL techniques.              |
| Final Python notebook                     | Final/review notebook with data-quality, KPI, SQL-verification and reproducibility checks.         |
| KPI documentation                         | KPI definitions, final values and interpretation documented in the project materials.              |
| Final business insights                   | Five evidence-based final business insights with recommended actions.                              |
| Recommendations                           | Management recommendations directly linked to validated analytical findings.                       |
| Validation evidence                       | Cross-tool validation plus Week 4 test/refine/re-test evidence.                                    |
| Data quality documentation                | Cleaning history plus final Week 4 data-quality confirmation.                                      |

| Required final Data Analytics deliverable | Final project output                                                     |
|-------------------------------------------|--------------------------------------------------------------------------|
| Final project documentation               | This consolidated final project report plus supporting Week 4 documents. |

**Final conclusion.** The FinTrust Data Analytics project progressed from business understanding and data preparation to advanced analysis, validation, dashboard refinement and final delivery. The final solution provides a coherent view of customer behaviour, transaction performance, operational failure patterns and synthetic risk-review activity. The major findings were tested across multiple tools, the final dashboard was functionally reviewed, and the limitations of the synthetic educational setting were documented. The project is therefore ready for final GitHub packaging and the individual 5–10-minute recorded presentation.